Topic to discuss

→ Richardson Extrapolation

→ Richardson with forward difference

→ Richardson with Backward difference

→ Richardson with central difference

Richardson Extrapolation

It is a technique used to improve the accuracy of a numerical approximation by combining results obtained with different step sizes.

We can estimate derivatives of a function using forward / backward difference but

- without Richardson, forward / backward are less accurate
- With Richardson, their accuracy improves
- Central difference + Richardson is usually best here

Richardson formula :

If an approximation $D(h)$ has error of order $O(h^n)$, then

$$D_{rich} = \frac{2^n D(h/2) - D(h)}{2^n - 1}$$

Another form, using h and rh, Instead of $h/2$, we reduce step size to rh, where $0 < r < 1$, then

$$D_{rich} = \frac{D(rh) - r^n D(h)}{1 - r^n}$$

This is the general form,

If $r = \frac{1}{2}$ then,

$$D_{rich} = \frac{D(rh) - r^n D(h)}{1 - r^n}$$

$$= \frac{D\left(\frac{h}{2}\right) - \left(\frac{1}{2}\right)^n D(h)}{1 - \left(\frac{1}{2}\right)^n}$$

· Multiply numerator and denominator by 2^n,

$$D_{rich} = \frac{2^n D\left(\frac{h}{2}\right) - D(h)}{2^n - 1}$$

How to choose 'n' value

It depends on the error order of the original method.

→ forward difference for first derivative :
error $O(h) \rightarrow n=1$

→ Backward difference for first derivative,
error $O(h) \rightarrow n=1$

→ Three-point central difference for first derivative,
error $O(h^2) \rightarrow n=2$

Numerical Problem :

Q: Estimate the value of $f'(x)$ at $x = 0.5$ using given data :

x	0	0.25	0.5	0.75	1.0
$f(x) = e^x$	1.0000	1.2840	1.6487	2.1170	2.7183

assume $h = 0.5$ and $r = 1/2$.

Solve it using

1) forward difference
2) Backward difference
3) Central difference / three point

solution : ① forward difference,

forward difference formula,

$$D(h) = \frac{f(x+h) - f(x)}{h}$$

for $n = 1$, and $r = \frac{1}{2}$,

$$D_{rich} = \frac{D(rh) - r^n D(h)}{1 - r^n}$$

$$D_{rich} = \frac{D\left(\frac{h}{2}\right) - \frac{1}{2} \times D(h)}{1 - \frac{1}{2}}$$

$$D_{rich} = 2D\left(\frac{h}{2}\right) - D(h)$$

Now, we have to find $D(h)$ & $D(h/2)$

We have, $h = 0.5$ & $h/2 = 0.25$

$$D(h) = \frac{f(x+h) - f(x)}{h} = \frac{f(0.5+0.5) - f(0.5)}{0.5}$$

$$= \frac{f(1) - f(0.5)}{0.5} = \frac{2.7183 - 1.6487}{0.5}$$

$$D(h) = 2.1392$$

and $$D\left(\frac{h}{2}\right) = \frac{f\left(x + \frac{h}{2}\right) - f(x)}{\frac{h}{2}} = \frac{f(0.75) - f(0.5)}{0.25}$$

$$= \frac{2.1170 - 1.6487}{0.25}$$

$$D\left(\frac{h}{2}\right) = 1.8732$$

So,
$$D_{rich} = 2D\left(\frac{h}{2}\right) - D(h)$$
$$= 2 \times 1.8732 - 2.1392$$
$$= 1.6072$$

③ Central difference / three point,

$$D(h) = \frac{f(x+h) - f(x-h)}{2h}$$

So, $n = 2$ and $r = \frac{1}{2}$

$$D_{rich} = \frac{D(rh) - r^n D(h)}{1 - r^n}$$

$$= \frac{D\left(\frac{h}{2}\right) - \frac{1}{4} D(h)}{1 - \frac{1}{4}}$$

$$D_{rich} = \frac{4 D\left(\frac{h}{2}\right) - D(h)}{3}$$

We have to find $D(h)$ & $D(h/2)$

also, $h = 0.5$ & $h/2 = 0.25$

$$D(h) = \frac{f(x+h) - f(x-h)}{2h}$$

$$= \frac{f(0.5+0.5) - f(0.5-0.5)}{2 \times 0.5}$$

$$= \frac{f(1) - f(0)}{1} = \frac{2.7183 - 1.0000}{1}$$

$$\therefore D(h) = 1.7183$$

now, $$D\left(\frac{h}{2}\right) = \frac{f\left(x+\frac{h}{2}\right) - f\left(x-\frac{h}{2}\right)}{\cancel{2} \times \frac{h}{\cancel{2}}}$$

$$= \frac{f(0.75) - f(0.25)}{0.5}$$

$$= \frac{2.1170 - 1.2840}{0.5} = 1.6660$$

Now,

$$D_{rich} = \frac{4D\left(\frac{h}{2}\right) - D(h)}{3}$$

$$= \frac{4 \times 1.6660 - 1.7183}{3}$$

$$= 1.6485667$$

Homework ② Solve using backward difference,

$$D(h) = \frac{f(x) - f(x-h)}{h}$$

$$D_{rich} = \frac{D(rh) - r\,D(h)}{1-r}$$

for $r = \frac{1}{2}$,

$$D_{rich} = 2D(h/2) - D(h)$$

Now, we have to find, $D(h)$ & $D(h/2)$

$$\therefore D(h) = \frac{f(0.5) - f(0.5-0.5)}{0.5}$$

$$\therefore D(h) = \frac{f(0.5) - f(0)}{0.5}$$

$$= \frac{1.6487 - 1.0000}{0.5}$$

$$= 1.2974$$

and $$D\left(\frac{h}{2}\right) = \frac{f(x) - f\left(x - \frac{h}{2}\right)}{h/2}$$

$$= \frac{f(0.5) - f(0.25)}{0.25}$$

$$= \frac{1.6487 - 1.2840}{0.25}$$

$$\therefore D(h/2) = 1.4588$$

Now,

$$D_{rich} = 2D(h/2) - D(h)$$

$$= 2 \times 1.4588 - 1.2974$$

$$= 1.6202$$

Ans.

Summary :

Forward + Richardson $\rightarrow$ 1.6072

Backward + Richardson $\rightarrow$ 1.6202

Central + Richardson $\rightarrow$ 1.6486

Now, exact solution,

$$f(x) = e^x$$

$$f'(x) = e^x$$

$$= e^{0.5} = 1.6487$$

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